

SAUMEN BISWAS

Senior / Staff SDET | Quality Engineering | Test Automation & Platform Quality

Python | Playwright | GraphQL / REST APIs | CI/CD | Cloud | AI-Assisted Quality

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PROFESSIONAL SUMMARY

Senior SDET / Quality Engineering leader with 18+ years of experience building test automation, validation platforms, release-confidence systems, and engineering-quality infrastructure for large-scale fintech, payments, SaaS, web, API, mobile, cloud-native, and distributed systems. Strong hands-on expertise in Python, Playwright, pytest, Selenium, GraphQL and REST API testing, JavaScript/TypeScript, SQL, CI/CD, AWS, Docker, and observability. Architected automation and quality intelligence used across 1,000+ repositories and 5,000-8,000 daily pull requests; delivered 99.5% data reliability, 60% faster CI feedback, 73% fewer flaky failures, and 94% lower mean time to detect issues. Experienced in AI-assisted test development, intelligent test selection, framework architecture, API/integration/E2E validation, release readiness, root-cause analysis, and technical leadership across high-reliability financial technology environments. Available immediately.

TECHNICAL EXPERTISE

Automation & Frameworks	Playwright, pytest, Selenium, Appium, XCTest, Espresso, Tosca; UI, API, integration, regression and end-to-end automation; reusable framework design, parallel execution, test-data design, cross-browser/device validation
API / Services / Data	Strong hands-on GraphQL testing; REST API automation, Postman, request/response and schema validation, service integration testing, SQL/backend validation, microservices and distributed-workflow validation
Programming	Python (strong), JavaScript/TypeScript, SQL, C#/.NET, Swift; Java familiarity
CI/CD & Infrastructure	Jenkins, GitHub Actions, Git, Docker, AWS, automated quality gates, pipeline optimization, test selection, release-readiness signals, cloud-native testing
AI-Assisted Quality Engineering	Claude, Cursor and LLM-assisted test generation, refactoring, failure summarization and triage; intelligent test selection/prioritization, self-healing automation, predictive quality metrics
Reliability / Observability	Datadog, Grafana, synthetic testing, telemetry, health scoring, flake detection, failure analysis, root-cause analysis, regression prevention, automated reporting
Quality Engineering & Leadership	Test strategy, risk-based testing, shift-left, TDD/BDD-oriented practices, test planning, defect triage, release readiness, quality metrics, mentoring, standards definition, cross-functional stakeholder communication
Domains	Fintech, payments, consumer lending, SaaS, large-scale distributed platforms, regulated/high-reliability systems

PROFESSIONAL EXPERIENCE

Senior Software Developer in Test / Technical Lead - Automation, Platform Quality & Reliability

Upstart Network, Inc. - San Mateo, CA | Apr 2022 - Aug 2026

- Architected and maintained scalable Python/pytest and Playwright automation spanning web UI, GraphQL and REST APIs, service integrations, SQL/data validation, and end-to-end workflows across a large-scale cloud-native fintech platform.
- Applied strong hands-on GraphQL experience to validate queries/mutations, request/response behavior, data integrity, error handling, and cross-service workflows alongside REST API automation.
- Built the Predictive Engineering Metrics Framework processing 5,000-8,000 pull requests daily across 1,000+ repositories, delivering 99.5% data reliability and 75% runtime reduction for engineering-quality analytics.

- Designed Smart Test Selection using impact analysis and intelligent prioritization, reducing average CI build times by 60% while preserving defect-detection signal and accelerating developer feedback.
- Championed AI-assisted quality engineering using Claude, Cursor and LLM workflows for test authoring, refactoring, failure summarization and triage; combined with reliability improvements to reduce flaky failures by 73%.
- Developed a proactive observability and staging-health framework using synthetic testing, automated health scoring and telemetry-driven signals, reducing mean time to detect issues by 94% and release delays by 78%.
- Integrated automated validation into Jenkins and GitHub Actions pipelines with release gates, parallel execution, flake detection, automated reporting and promotion-readiness signals.
- Performed API, integration, SQL/data and cross-system validation across distributed workflows; reproduced production issues, led root-cause investigations and converted defects into durable regression coverage.
- Built reusable automation components, utilities, assertions and test-data patterns to improve maintainability, deterministic execution and multi-environment scale across engineering teams.
- Partnered with Product, Engineering, Platform and Infrastructure teams on acceptance criteria, testability, shift-left risk identification, defect triage and release readiness; mentored engineers on automation architecture and production-quality test code.
- Led AWS cloud reliability and cost-governance analysis across large-scale infrastructure, delivering \$2.24M in annual savings while protecting platform performance and reliability.

Senior Software Engineer - Quality Engineering

Teladoc Health - Mountain View, CA | Nov 2020 - Mar 2022

- Built and improved automation and validation for large-scale SaaS applications across UI, API, integration, regression and end-to-end layers; partnered with development teams on testability and release quality.
- Increased automated test coverage by 30% and reduced post-release defects by 25% through disciplined test design, coverage analysis, CI-oriented quality practices and defect prevention.
- Developed and maintained iOS automated tests using XCTest and Swift in Xcode, improving functional/component coverage and strengthening device-facing validation experience.
- Audited and stabilized automated suites; debugged flaky, intermittent and asynchronous failures and improved deterministic execution in CI.
- Contributed to end-to-end testing using Tosca, API/integration validation, defect triage, release readiness and continuous improvement in a fast-paced Agile environment.

Senior Staff Software Engineer - Quality & Release Engineering

Visa Inc. - Foster City, CA | Dec 2014 - Oct 2020

- Led end-to-end quality engineering and release validation for high-scale global payment and tokenization systems, including the Apple Pay launch, coordinating complex cross-system validation and achieving zero critical production defects on major releases.
- Designed TWIK, a hybrid rule-based and statistical test-prioritization system that reduced regression execution time by 25-40% while improving defect-detection efficiency for transaction-processing workflows.
- Built Auth24x7, a continuous regression framework for 24x7 payment processing, strengthening reliability and reducing operational errors and related costs.
- Owned integration strategy, release-readiness criteria, quality gates, cross-system defect analysis and production-risk assessment across distributed engineering teams in a regulated environment.
- Established reusable automation standards and scalable UI/API/integration validation approaches; mentored engineers and drove root-cause analysis and regression prevention across teams.

- Supported high-volume transaction and payment-processing validation, including data integrity, service interoperability, performance/capacity readiness and complex client/network integrations.
- Recognized with Visa Go Beyond Innovation Awards in 2017 and 2019 for impactful engineering, automation and quality contributions.

Senior Software Engineer

R S Software (India) Ltd. - Kolkata, India | Mar 2007 - Nov 2014

- Delivered full-lifecycle software engineering and test automation for enterprise and payment systems, including requirements analysis, implementation, system/integration testing, defect resolution and production release support.
- Built automation and data-driven components using C#/.NET and SQL, establishing a strong software-engineering foundation for scalable validation systems and backend data verification.
- Worked in services/consulting delivery models with onshore and offshore stakeholders, adapting quickly to evolving client requirements, technologies, schedules and quality expectations.

SELECTED PROJECT DETAILS

- **Automation Platform - Web / API / GraphQL:** Designed and maintained Python/pytest and Playwright automation across UI, GraphQL, REST APIs, service integrations and end-to-end flows. Built reusable components, assertions, data utilities and CI execution patterns for maintainable multi-team adoption.
- **Smart Test Selection / AI-Assisted QE:** Implemented impact-based test prioritization and AI-assisted test development/analysis, reducing CI feedback time by 60% and contributing to a 73% reduction in flaky failures.
- **Global Payments Automation - Visa:** Led quality engineering for high-volume payment/tokenization programs including Apple Pay; created TWIK test prioritization and Auth24x7 continuous regression capabilities for mission-critical transaction processing.

SELECTED ENGINEERING IMPACT

- **Enterprise-Scale Predictive Engineering Metrics:** Built engineering-quality analytics across 1,000+ repositories and 5,000-8,000 daily pull requests; achieved 99.5% data reliability and 75% processing-runtime reduction.
- **Intelligent Test Selection & CI Optimization:** Applied impact analysis and intelligent prioritization to reduce CI feedback time by 60%; reliability work reduced flaky failures by 73%.
- **Global Payments Quality:** Led quality/release validation for mission-critical Visa payment initiatives including Apple Pay; developed TWIK and Auth24x7 automation approaches for high-volume transaction systems.

CONSULTING / CLIENT DELIVERY STRENGTHS

- **Automation Architecture:** Hands-on design of reusable Python/Playwright/pytest automation, API and GraphQL validation, data assertions, CI execution, parallelization, stability patterns and maintainable test infrastructure.
- **Client-Facing Quality Leadership:** Experienced communicating technical risk, defect trends, quality metrics and go/no-go readiness to engineering, product and business stakeholders while mentoring distributed teams.
- **Cross-Platform Coverage:** Web, API, backend/data, cloud-native and mobile validation using Playwright, Selenium, Appium, XCTest/Swift, Espresso exposure and Tosca, supporting broad SDET/QE client requirements.

EDUCATION, CERTIFICATIONS & PROFESSIONAL RECOGNITION

- M.S., Information Technology Management - Western Governors University, 2020
- B.S., Electrical Engineering - West Bengal University of Technology, 2005
- CAPM (PMI) | Certified ScrumMaster (CSM) | Certified Tester Foundation Level (CTFL) | Machine Learning - Stanford Coursera
- IEEE Senior Member | Visa Go Beyond Innovation Award (2017, 2019)